Machine Learning Pipeline

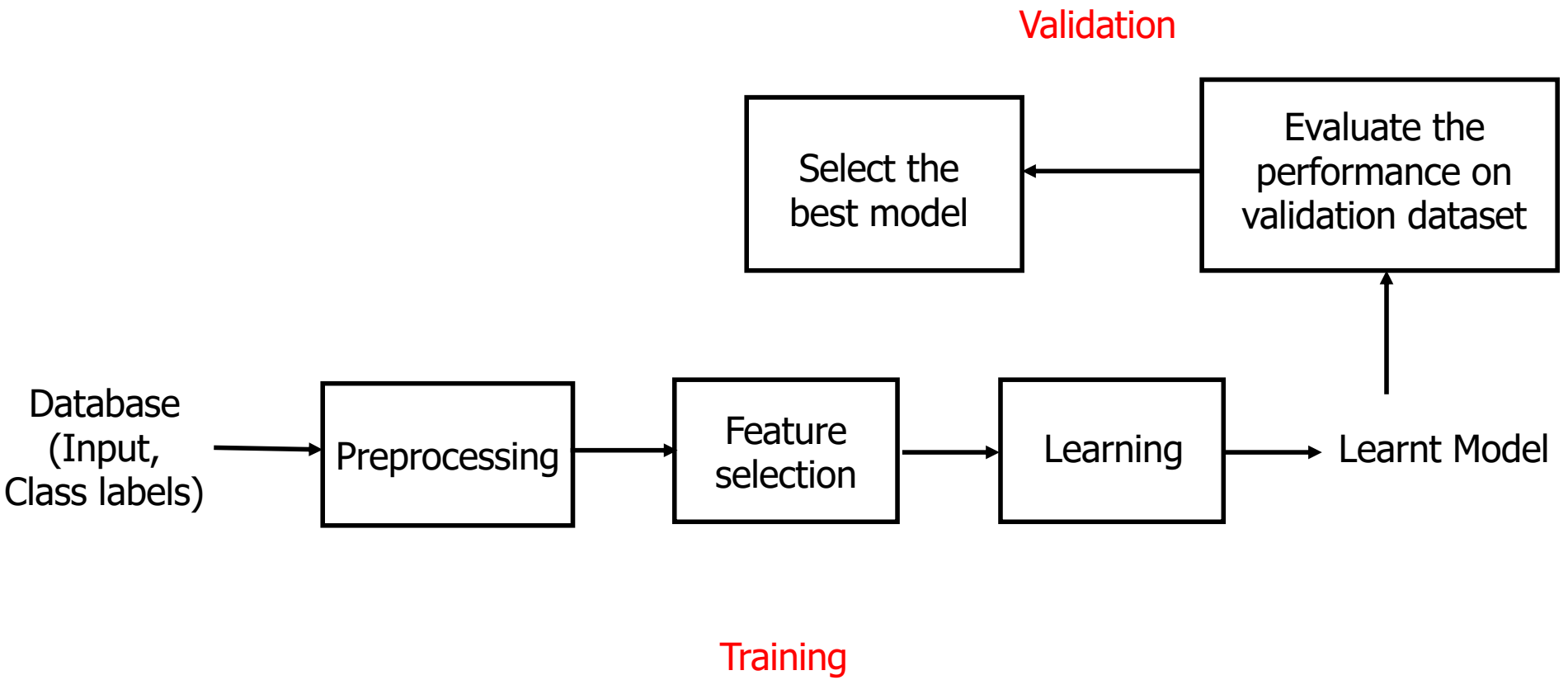
- Three steps:
  - Training
  - Validation
  - Testing

Machine Learning Pipeline

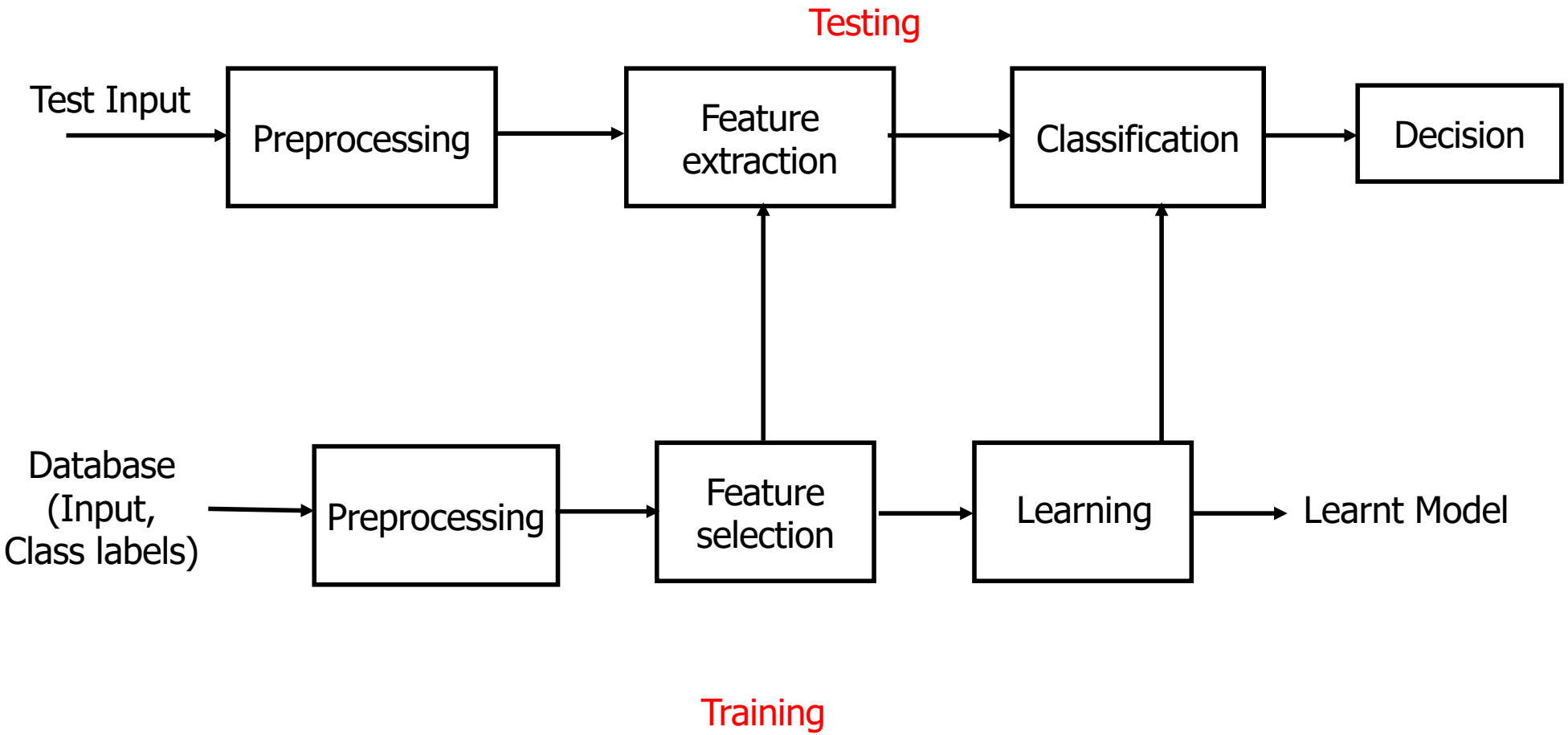


Training

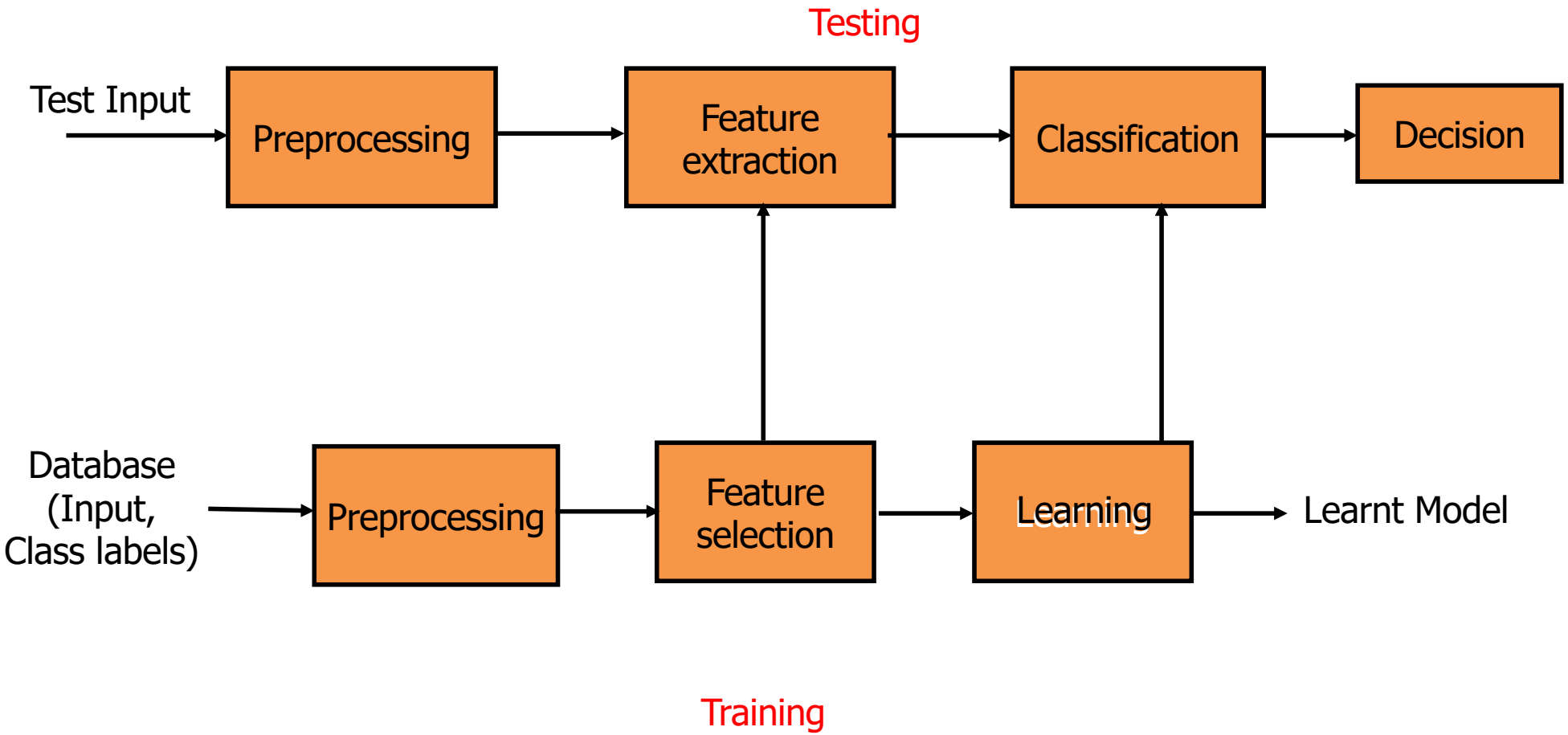
Machine Learning Pipeline



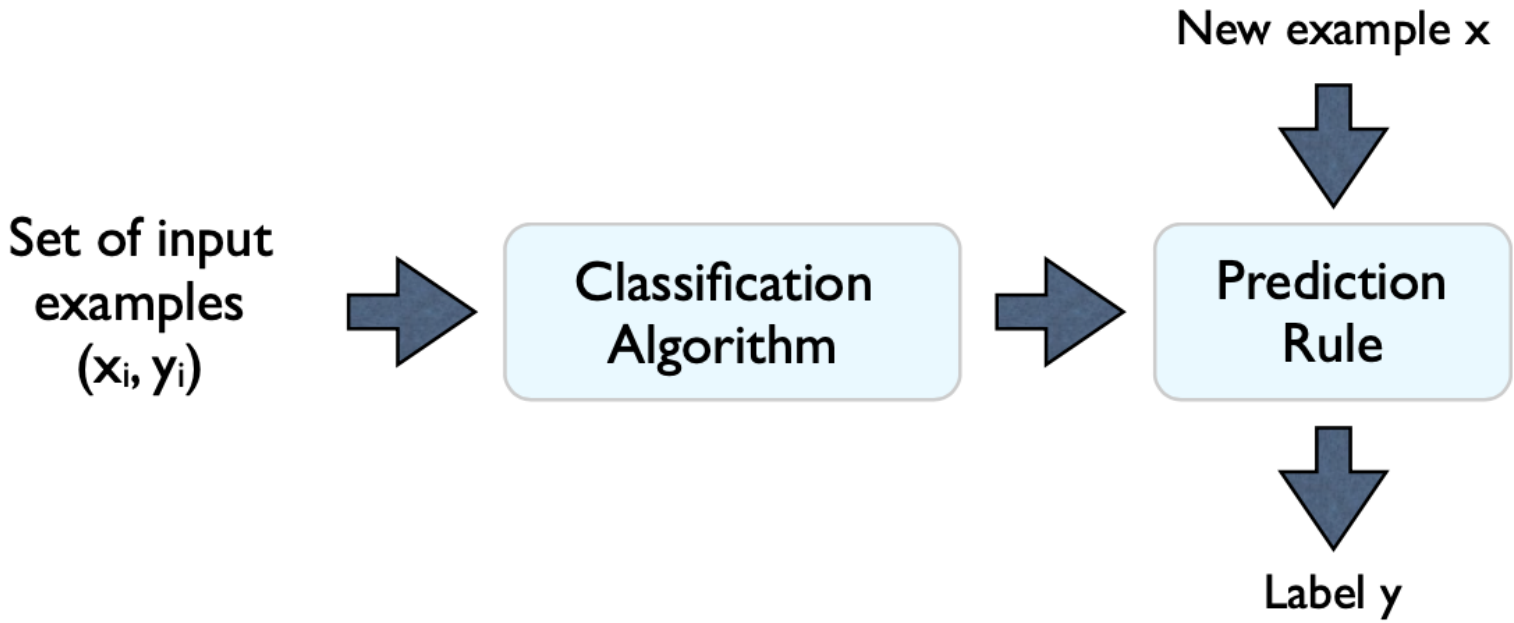
Machine Learning Pipeline



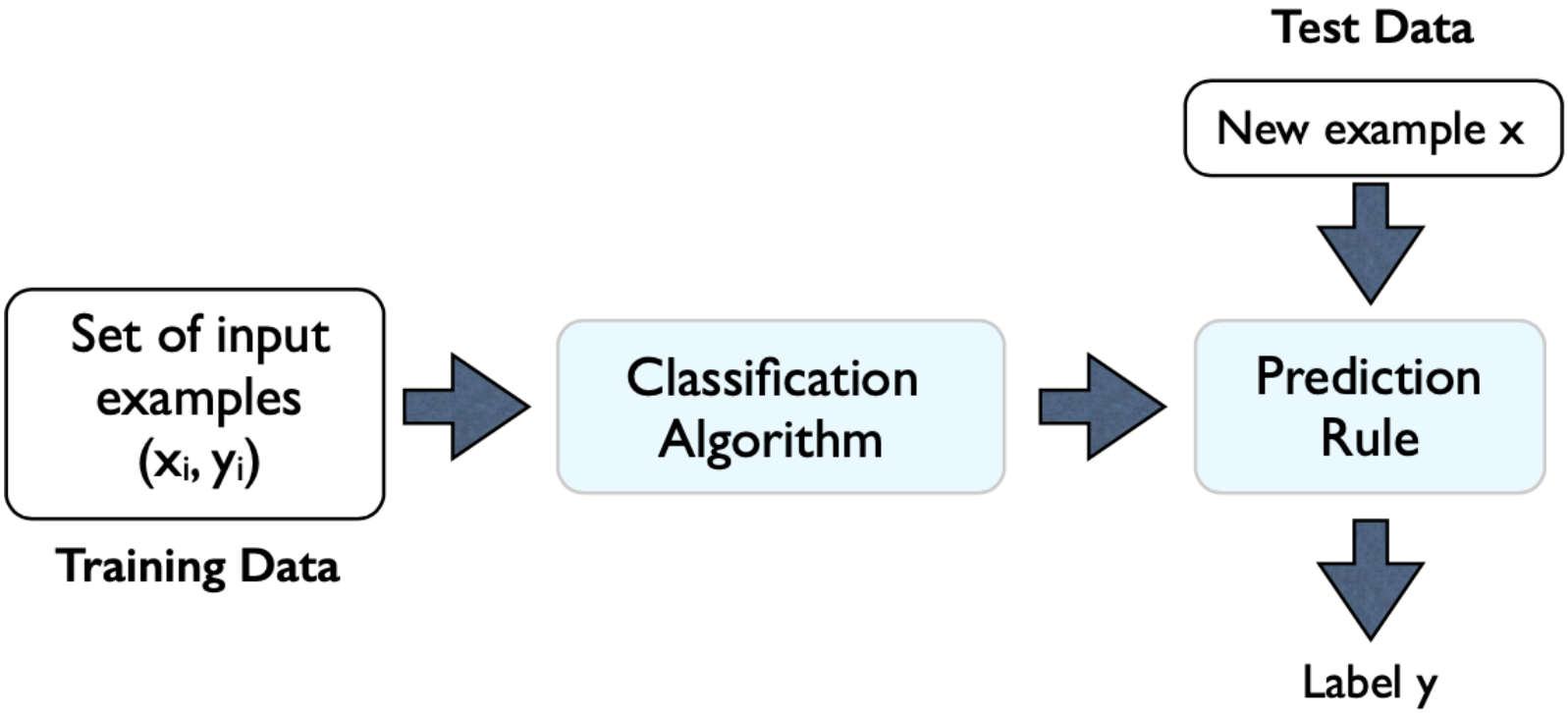
Machine Learning Pipeline



Typical Classification Algorithm



Typical Classification Algorithm



Training and test data must be **separate**!

Performance Measure:
Accuracy (or fraction of correct answers) on **test data**

ML Paradigms and Evaluation Metrics

Types of ML Paradigms

- Supervised learning
  - Classification
  - Regression
- Unsupervised learning
- Reinforcement learning

Supervised Learning

- Given: training data + desired outputs (labels)
- (x1, y1), (x2, y2), ..., (xn, yn)
- Learn a function f(x) to predict y given x

Cats



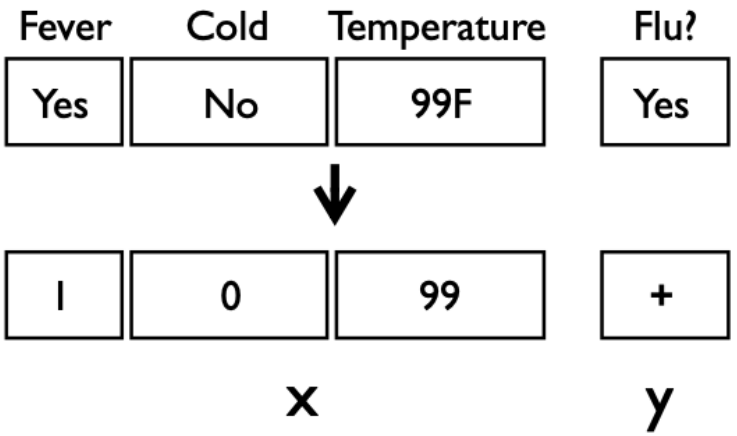


Dogs

Supervised Learning

Example 1: Predict if a **new** patient has flu or not, based on **existing** patient data

What is x and y?



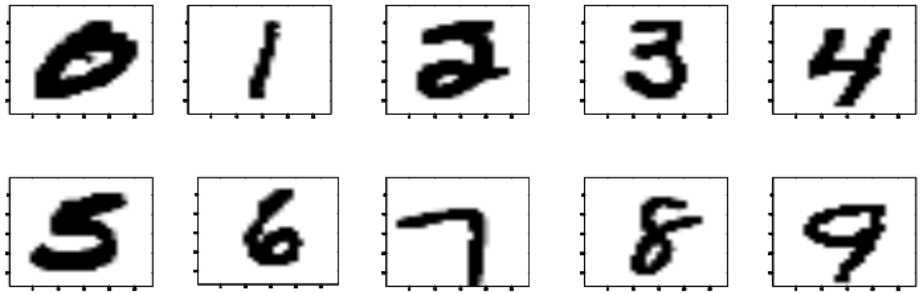
Features: Properties of patient

Label: Flu/No flu

A **binary** (two-label) classification problem

Supervised Learning

Example 2: Which digit in the image ?

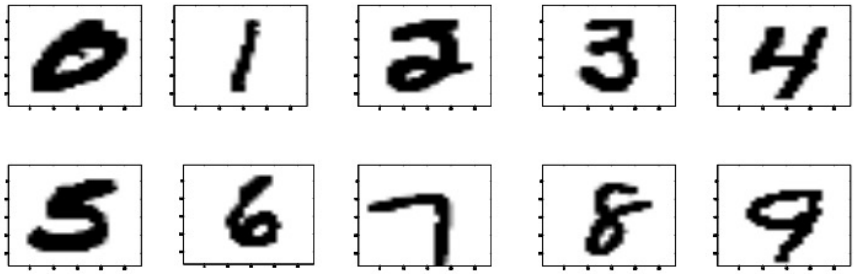


Label: 0,1,..,9

What are the features?

A **multiclass** classification problem

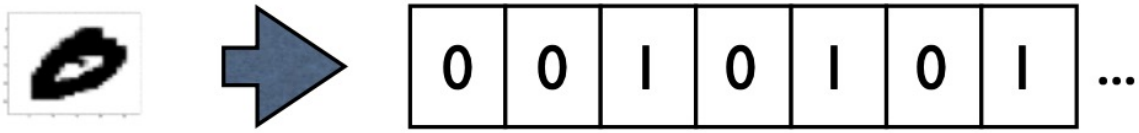
Supervised Learning



Label: 0,1,..,9

What are the features?

Option: vector of pixel colors



Image

x (0 for white, 1 for black)

There are other options too

Lesson: Choosing features is non-trivial in real applications

Supervised Learning

Example 3: Spam or not?

Email1
From: Canadian Pharmacy
Subject: Offer ends now!

Email 2
From: Yuncong Chen
Subject: TA meeting

	Pharmacy	offer	meeting	TA	Spam?
Email 1	1	1	0	0	Yes
Email 2	0	0	1	1	No

Label: 0 (not spam), 1 (spam)

Features: Words in the email

Supervised Learning

Regression:

Given data:

$$(x_i, \quad y_i) \qquad i=1,..,n$$

independent variable

dependent variable

where y is **continuous**, design a rule to predict y values for unseen x

Supervised Learning

Regression: Given data (x_i, y_i)
where y is **continuous**, predict y values for unseen x

Example 1: Predict house price from properties of house

Bedrooms	Bathrooms	Area	Price
3	2	2000	600K
2	1	1200	400K
	x		y

Independent Variable: Property of house

Dependent variable: price

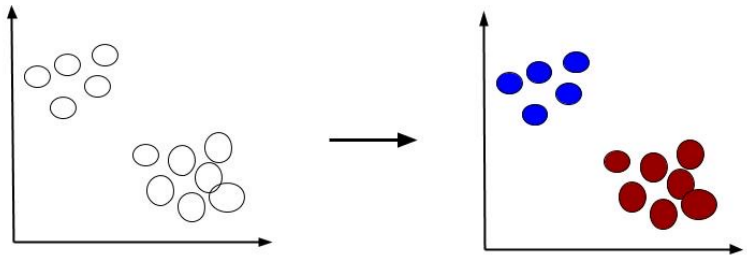
Supervised Learning



Unsupervised Learning

- Given: training data (without desired outputs)
- x1, x2, ..., xn (without labels)
- Output hidden structure behind the x's – Group them by similarity





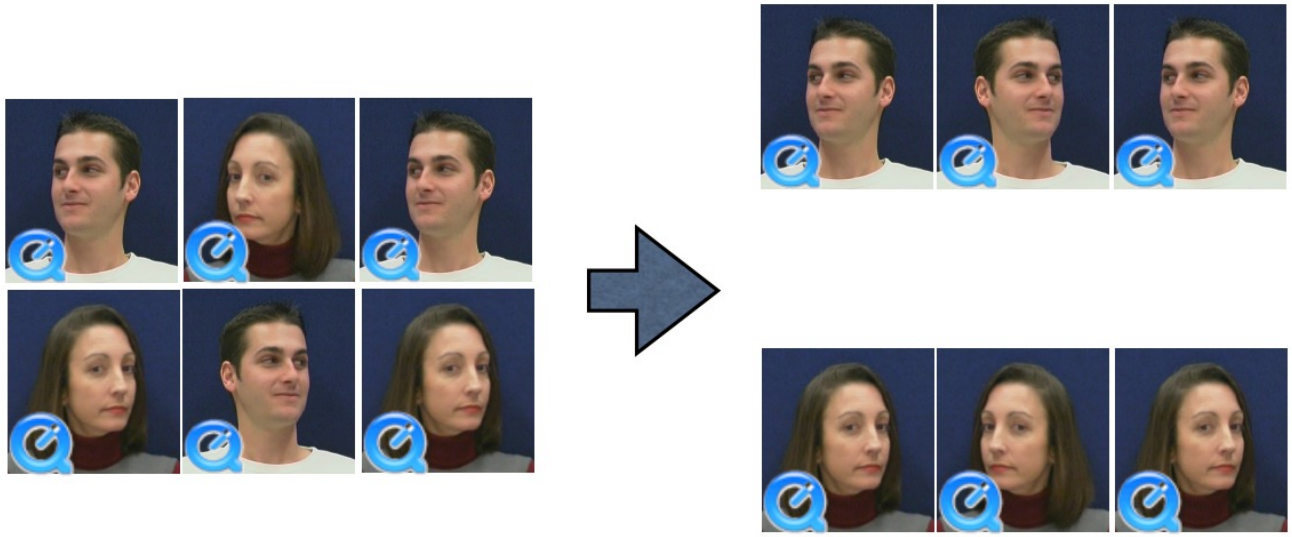
Clustering

Unsupervised Learning

Clustering

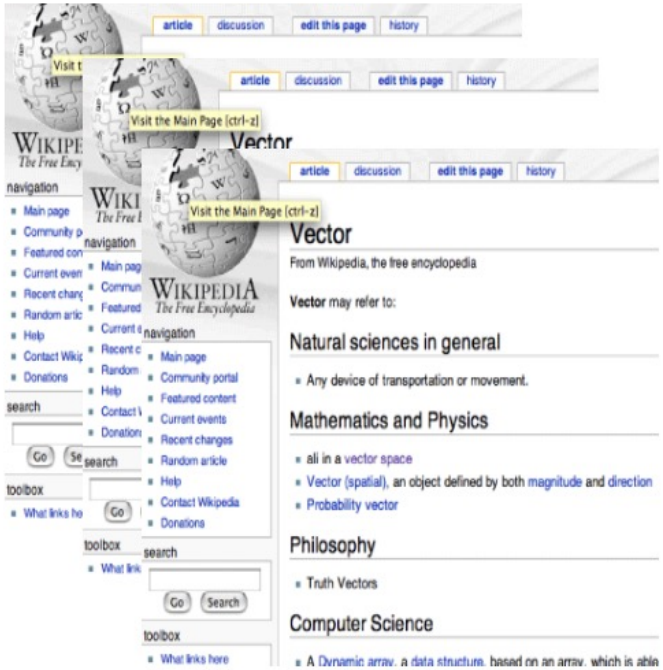
Given a set of input objects, group them to clusters by similarity

Example 1: Cluster videos by people in them



Unsupervised Learning

Example 2: Cluster documents by topic



Physics

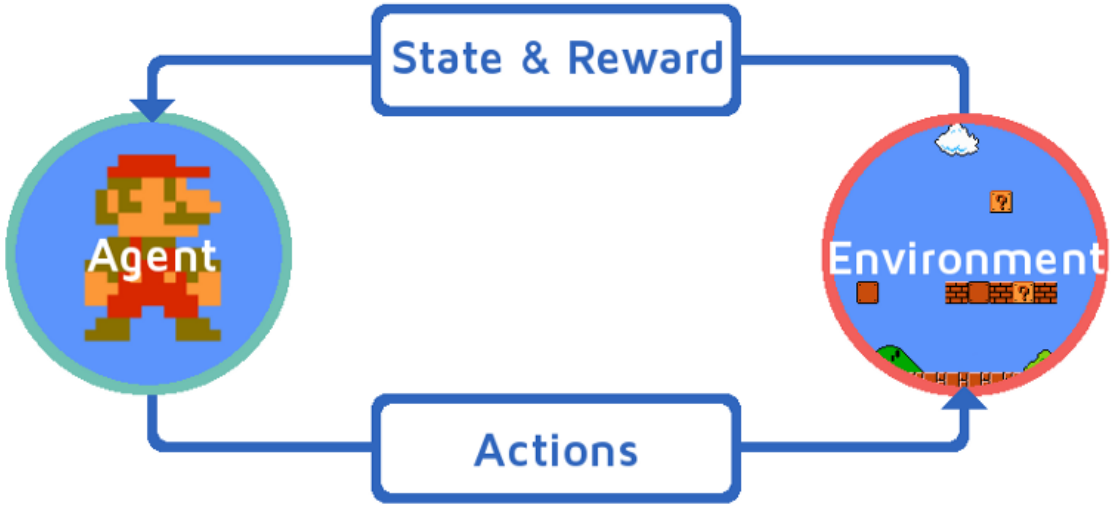
- Gravity
- Laws of Motion
- Electricity

Math

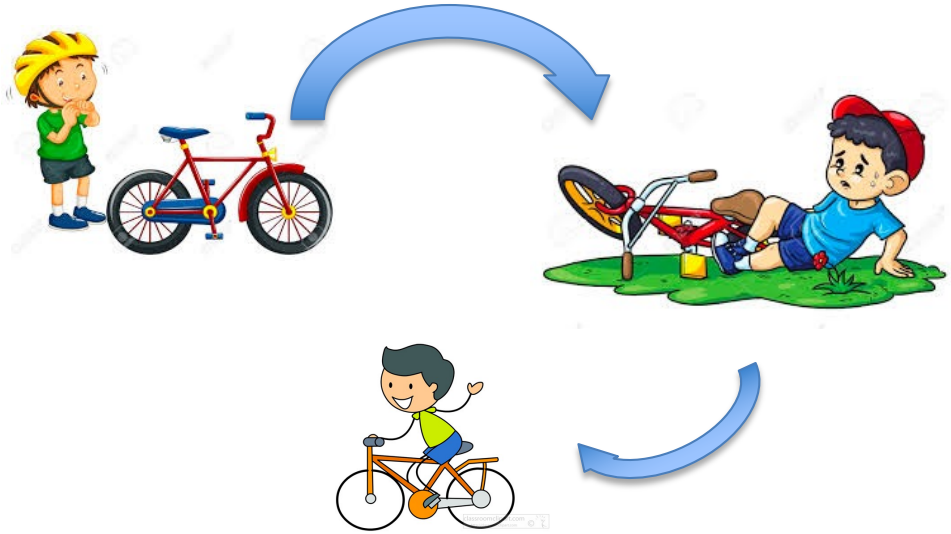
- Geometry
- Algebra

Features: Words in the document

Reinforcement Learning



Rewards from sequence of actions



Reinforcement Learning

- Playing games: Atari, Chess, Checkers
- Robot navigation
- Talent acquisition

Applications of Learning Paradigms

Supervised

- Person identification
- Object recognition
- Stock prediction

Reinforcement

- Game playing
- Credit assignment

Unsupervised

- Social network analysis
- Dimensionality reduction
- Market segmentation

More Recently…

- Combination of these paradigms are being explored:
  - Semi-supervised learning
  - Self supervised learning
  - Supervised + reinforcement
  - ….

A PROPOSAL FOR THE DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

1956 Dartmouth Conference: The Founding Fathers of AI



John MacCarthy | Marvin Minsky | Claude Shannon | Ray Solomonoff | Alan Newell

Herbert Simon | Arthur Samuel | Oliver Selfridge | Nathaniel Rochester | Trenchard More

J. McCarthy, Dartmouth College
M. L. Minsky, Harvard University
N. Rochester, I.B.M. Corporation
C.E. Shannon, Bell Telephone Laboratories

1 **Automatic Computers**

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

2. **How Can a Computer be Programmed to Use a Language**

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning and rules of conjecture. From this point of view, forming a generalization consists of admitting a new word and some rules whereby sentences containing it imply and are implied by others. This idea has never been very precisely formulated nor have examples been worked out.

3. **Neuron Nets**

How can a set of (hypothetical) neurons be arranged so as to form concepts. Considerable theoretical and experimental work has been done on this problem by Uttley, Rashevsky and his group, Farley and Clark, Pitts and McCulloch, Minsky, Rochester and Holland, and others. Partial results have been obtained but the problem needs more theoretical work.

4. **Theory of the Size of a Calculation**

If we are given a well-defined problem (one for which it is possible to test mechanically whether or not a proposed answer is a valid answer) one way of solving it is to try all possible answers in order. This method is inefficient, and to exclude it one must have some criterion for efficiency of calculation. Some consideration will show that to get a measure of the efficiency of a calculation it is necessary to have on hand a method of measuring the complexity of calculating devices which in turn can be done if one has a theory of the complexity of functions. Some partial results on this problem have been obtained by Shannon, and also by McCarthy.

5. **Self-Improvement**

Probably a truly intelligent machine will carry out activities which may best be described as self-improvement. Some schemes for doing this have been proposed and are worth further study. It seems likely that this question can be studied abstractly as well.

6. **Abstractions**

A number of types of "abstraction" can be distinctly defined and several others less distinctly. A direct attempt to classify these and to describe machine methods of forming abstractions from sensory and other data would seem worthwhile.

7. **Randomness and Creativity**

A fairly attractive and yet clearly incomplete conjecture is that the difference between creative thinking and unimaginative competent thinking lies in the injection of a some randomness. The randomness must be guided by intuition to be efficient. In other words, the educated guess or the hunch include controlled randomness in otherwise orderly thinking.

History of ML

- 1950s
  - Samuel's checker player
  - Selfridge's Pandemonium
- 1960s:
  - Neural networks: Perceptron
  - Pattern recognition
  - Learning in the limit theory
  - Minsky and Papert prove limitations of Perceptron
- 1970s:
  - Symbolic concept induction
  - Winston's arch learner
  - Expert systems and the knowledge acquisition bottleneck
  - Quinlan's ID3
  - Michalski's AQ and soybean diagnosis
  - Scientific discovery with BACON
  - Mathematical discovery with AM

History of ML

- 2000s
  - Support vector machines & kernel methods
  - Graphical models
  - Statistical relational learning
  - Transfer learning
  - Sequence labeling
  - Collective classification and structured outputs
  - Computer Systems Applications (Compilers, Debugging, Graphics, Security)
  - E-mail management
  - Personalized assistants that learn
  - Learning in robotics and vision
- 2010s
  - Deep learning systems
  - Learning for big data
  - Bayesian methods
  - Multi-task & lifelong learning
  - Applications to vision, speech, social networks, learning to read, etc.
- 2020s
  - Deep learning
  - Foundation Models

Based on slide by Ray Mooney

Evaluation Metrics:

Let us design a simple classification algorithm

Purse vs Laptop Bag: Design a classifier










Laptop bag vs. Purse: Design a classifier

- Features:
  - Width
  - Height
  - Weight
  - Embeddings

- Classifier: threshold/MLP/loss function

Evaluation Metrics

Let the problem statement be *classifying purse and bags*.
Purses are labeled as positive class and bags are labeled as negative class

		Predicted Class	
		Negative	**Positive**
Actual Class	**Negative**	A (true negative)	C (false positive)
	Positive	D (false negative)	B (true positive)

Term	Meaning	Example
True positive	Correct classification	Purse identified as purse
False positive	Incorrect classification	Bag identified as purse
True negative	Correct classification	Bag identified as bag
False negative	Incorrect classification	Purse identified as bag

http://www.cs.rpi.edu/~leen/misc-publications/SomeStatDefs.html

Evaluation Metrics

		Predicted Class	
		Negative	**Positive**
Actual Class	**Negative - 50**	A (true negative) - 40	C (false positive) - 10
	Positive - 50	D (false negative) - 20	B (true positive) - 30

		Negative	**Positive**
Actual Class	**Negative - 95**	A (true negative) - 90	C (false positive) - 5
	Positive - 5	D (false negative) - 3	B (true positive) - 2

Metric	Formula
Average classification accuracy	[(TN + TP) / (TN + FP + TP + FN)]
Class-wise classification accuracy	[TN / (TN + FP) + TP / (TP + FN)]/2
Type I error (false positive rate)	FP / (TN + FP)
Type II error (false negative rate)	FN / (FN + TP)
True positive rate	TP / (TP + FN)
True negative rate	TN / (TN + FP)

http://www.cs.rpi.edu/~leen/misc-publications/SomeStatDefs.html

Evaluation Metrics

Metric	Formula
Average classification accuracy	(TN + TP) / (TN+TP+FN+FP)
Type I error (false positive rate)	FP / (TN + FP)
Type II error (false negative rate)	FN / (FN + TP)
True positive rate	TP / (TP + FN)
True negative rate	TN / (TN + FP)

- Type I error or false positive rate: The chance of incorrectly classifying a (randomly selected) sample as positive
- Type II error or false negative rate: The chance of incorrectly classification a (randomly selected) sample as negative

Evaluation Metrics

Metric	Formula
Average classification accuracy	(TN + TP) / (TN+TP+FN+FP)
Type I error (false positive rate)	FP / (TN + FP)
Type II error (false negative rate)	FN / (FN + TP)
True positive rate	TP / (TP + FN)
True negative rate	TN / (TN + FP)

- Type I error or false positive rate: The chance of incorrectly classifying a (randomly selected) sample as positiv
- Type II error or false negative rate: The cha classification a (randomly selected) samp

Prevalent in computer vision and image processing related classification problems